DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Assignment Rubric & Specification

Course Code:	CS201
Course Name:	Data Structures & Algorithms
Assignment Title:	Assignment 4 - Red-Black Tree Implementation
Submission Deadline:	2024-12-05 23:59:00
Maximum Marks:	100 Marks
Instructor:	Dr. Meena Iyer

1. Objective & Problem Statement

Students are required to implement a self-balancing Red-Black Search Tree in C++ or Java. The implementation must support node insertion, deletion, left/right rotations, color flips, and verification of tree properties. Benchmark time complexity against unbalanced BSTs.

2. Evaluation Criteria & Marking Rubric

Component	Description	Marks
Node Insertion & Rotations	Correct tree rebalancing with left and right rotations.	30%
Node Deletion & Fixup	Handling double-black conditions and coloring fixes.	30%
Performance Benchmarking	Empirical time complexity graphs for N=10,000 nodes.	20%
Code Quality & Tests	Clean modular architecture and unit test suite.	20%

3. Submission Guidelines

Upload a compressed ZIP archive containing source code, README, and PDF benchmark report. Strict penalty applies for late submissions. Academic integrity and plagiarism policies are strictly enforced.